

T-11

$$H_0: \xi \sim p_0(x) = \mathcal{U}(0; 1)$$

$$H_1: \xi \sim p_1(x) = \frac{e^{1-x}}{e-1} \mathcal{U}(0; 1)$$

a)  $n=1$  —  $\text{yp. } \text{гар.}-\text{тн}$   $\alpha$

$$l(x) = \frac{L_1}{L_0} = \frac{p_1(x)}{p_0(x)} = \frac{e^{1-x}}{e-1} \geq C$$

$$G: x \leq A$$

$$P(x \in A | H_0) = \alpha$$

$$\int_0^A dx = \alpha \Rightarrow A = \alpha \Rightarrow G: x \leq \alpha$$

$$W = P(x \in A | H_1) = \int_0^A \frac{e^{1-x}}{e-1} dx = \frac{e}{e-1} \cdot \frac{e^{-x}}{(-1)} \Big|_0^A = \frac{e}{e-1} (1 - e^{-\alpha})$$

$$\alpha_1 = \alpha$$

$$\alpha_2 = 1 - W$$

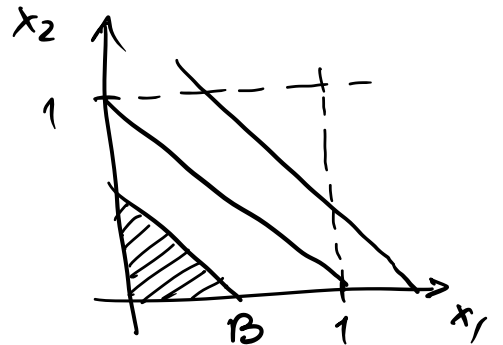
б)  $n=2$

$$l = \frac{L_1}{L_0} = \frac{\frac{e^{1-x_1}}{e-1} \cdot \frac{e^{1-x_2}}{e-1}}{1 \cdot 1} = \frac{e^{2-x_1-x_2}}{(e-1)^2} \geq C$$

$$G: x_1 + x_2 \leq B$$

$$P(x_1 + x_2 \leq B | H_0) = \alpha$$

$$\iint_{x_1+x_2 \leq B} dx_1 dx_2 = \alpha$$
$$\frac{B^2}{2} = \alpha \Rightarrow B = \sqrt{2\alpha}$$
$$B \leq 1$$



$$\sqrt{2\alpha} < 1 \Rightarrow \alpha < \frac{1}{2}$$

$$G: x_1 + x_2 \leq B = \sqrt{2\alpha}$$

$$W = P(x_1 + x_2 \leq B | H_1) = \iint_{\substack{x_1+x_2 \leq B \\ 0 \leq x_i < 1}} \frac{e^{1-x_1} \cdot e^{1-x_2}}{(e-1)^2} dx_1 dx_2 =$$

$$= \frac{e^2}{(e-1)^2} \int_0^B dx_1 \int_0^{B-x_1} e^{-x_1} \cdot e^{-x_2} dx_2 = \frac{e}{(e-1)^2} \int_0^B e^{-x_1} (1 - e^{-B+x_1}) dx_1 =$$

$$= \frac{e^2}{(e-1)^2} [1 - e^{-B} - e^{-B} \cdot B], \text{ где } B = \sqrt{2\alpha}$$

$$\begin{cases} \alpha_1 = \alpha \\ \alpha_2 = 1 - \alpha \end{cases}$$

б) Асимптот. критерий по выборке  $n$

$$L(\vec{x}_n) = \frac{L_1(\vec{x}_n)}{L_0(\vec{x}_n)} = \frac{\prod_{i=1}^n p_i(x_i)}{\prod_{i=1}^n p_0(x_i)} = \prod_{i=1}^n \frac{p_i(x_i)}{p_0(x_i)} \geq e$$

$$\ln L = \sum_{i=1}^n \underbrace{\ln \left( \frac{p_i(x_i)}{p_0(x_i)} \right)}_{\gamma_i} \geq \ln e \Rightarrow \sum_{i=1}^n \ln p_i(x_i) - \sum_{i=1}^n \ln p_0(x_i) =$$

$$= \sum_{i=1}^n \ln \frac{e^{1-x}}{e-1} - 0 \geq \ln e$$

$$\sum_{i=1}^n (1-x_i) - n \ln(e-1) \geq \ln e$$

$$G: \sum_{i=1}^n x_i \leq A \mid : n \Rightarrow \bar{x} \in B; \quad P(\bar{x} \leq B \mid H_0) = \alpha$$

УПТ

$$\frac{\bar{x} - \mu_{\xi}}{\sqrt{\sigma_{\xi}}} \sqrt{n} \rightsquigarrow N(0, 1)$$

$$H_0: \mu_{\xi} = \frac{1}{2}; \quad \sigma_{\xi} = \frac{1}{\sqrt{12}}; \quad P(\bar{x} \leq B \mid H_0) = P\left(\frac{\bar{x} - \frac{1}{2}}{\sqrt{\frac{1}{12}}} \sqrt{n} \leq \frac{B - \frac{1}{2}}{\sqrt{\frac{1}{12}}} \sqrt{n}\right)$$

$$u_{\alpha}$$

$$B = \frac{1}{2} + \frac{u_{\alpha}}{\sqrt{12n}}$$

$$\alpha = P(\bar{x} \leq B \mid H_1)$$

$$H_1: \mu_{\xi} = \int_0^1 x p_1(x) dx = \frac{e-2}{e-1} \quad \mu_{\xi}^2 = \int_0^1 x^2 p_1(x) dx = \frac{2e-5}{e-1}$$

$$\sigma_{\xi}^2 = \frac{e^2 - 3e + 1}{(e-1)^2}$$

$$P\left(\frac{\bar{x} - \mu_{\xi}}{\sqrt{\sigma_{\xi}^2}} \leq \frac{B - \mu_{\xi}}{\sqrt{\sigma_{\xi}^2}} \sqrt{n}\right) = \int_{-\infty}^{\delta} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx =$$

$$= \frac{\frac{\mu_2}{\sqrt{12}} + 0,082\sqrt{n}}{\sqrt{\sigma_{\xi}^2}} \xrightarrow{n \rightarrow \infty} +\infty \quad \int \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \xrightarrow{n \rightarrow \infty} 1$$

$$\Rightarrow W \xrightarrow{n \rightarrow \infty} 1 \quad \begin{cases} \alpha_1 = 1 \\ \alpha_2 = 1 - \alpha \end{cases}$$

2) Построить опт. критерий по выборке объема  $n$  с крит. обл.  $G: x_{\min} \leq C$  и ур-ю знач-ти  $\alpha$ .

$$\alpha_2 \rightarrow \min \quad \alpha_1 = P(x_{\min} \leq C | H_0) = 1 - (1 - f(C))^n \leq \alpha$$

$$\alpha_1 \leq \alpha \quad 1 - (1 - C)^n \leq \alpha \Rightarrow 1 - \sqrt[n]{1 - \alpha} \geq C$$

$$\alpha_2 = P(x_{\min} > C | H_1) = 1 - P(x_{\min} \leq C | H_1) = (1 - \Phi(C))^n =$$

$$= \left\{ \Phi(C) = \frac{e}{e-1} (1 - e^{-C}) \right\}^n = \left( 1 - \frac{e}{e-1} (1 - e^{-C}) \right)^n$$

$$\frac{d\alpha_2}{dC} = n \left( 1 - \frac{e}{e-1} (1 - e^{-C}) \right)^{n-1} \left( \frac{-e^{1-C}}{e-1} \right) = 0$$

$$\frac{e}{e-1} (1 - e^{-C}) = 1$$

$$e^{-C} = e^{-1} \Rightarrow C = 1$$

$$\alpha_1 \leq \alpha \Rightarrow 1 - \sqrt[n]{1 - \alpha} \geq C$$

$$1 - \alpha \in (0; 1) \Rightarrow$$

$$\Rightarrow \sqrt[n]{1 - \alpha} \in (0; 1) \Rightarrow$$

$$\Rightarrow 1 - \sqrt[n]{1 - \alpha} \in (0; 1) \Rightarrow$$

$$\Rightarrow C > 1 - \sqrt[n]{1 - \alpha} \text{ (yes, good } \alpha_2 \text{ ✓)}$$

$$\frac{d^2 \alpha_2}{dc^2} = n(n-1) \left( 1 - \frac{e}{e-1} (1 - e^{-c}) \right)^{n-2} \left( -\frac{e^{1-c}}{e-1} \right)^2 + n \left( 1 - \frac{e}{e-1} (1 - e^{-c}) \right)^{n-1} \cdot \left( \frac{e^{1-c}}{e-1} \right)$$

> 0

$$\Rightarrow C = \gamma \cdot \min$$

$$C \leq 1 - \sqrt[n]{1-\alpha} \Rightarrow C = 1 - \sqrt[n]{1-\alpha}$$

$\alpha_1 = \alpha$

$$\Rightarrow G: X_{\min} < 1 - \sqrt[n]{1-\alpha}$$

$$W = P(X_{\min} < 1 - \sqrt[n]{1-\alpha} | H_1) = 1 - (1 - \Phi(e))^n =$$

$$= 1 - \left( 1 - \frac{e}{e-1} (1 - e^{-c}) \right)^n = 1 - \left( 1 - \frac{e}{e-1} (1 - e^{-1 + \sqrt[n]{1-\alpha}}) \right)^n \quad \textcircled{=}$$

$$-1 + \sqrt[n]{1-\alpha} = -1 + e^{\frac{1}{n} \ln(1-\alpha)} = -1 + 1 + \frac{1}{n} \ln(1-\alpha) + O\left(\frac{1}{n^2}\right) =$$

$$= \frac{1}{n} \ln(1-\alpha) + O\left(\frac{1}{n^2}\right)$$

$$e^{-1 + \sqrt[n]{1-\alpha}} = e^{\frac{1}{n} \ln(1-\alpha) + O\left(\frac{1}{n^2}\right)} = 1 + \frac{1}{n} \ln(1-\alpha) + O\left(\frac{1}{n^2}\right)$$

$$\textcircled{=} 1 - \left( 1 - \frac{e}{e-1} \left( 1 - 1 + \frac{1}{n} \ln(1-\alpha) + O\left(\frac{1}{n^2}\right) \right) \right)^n = 1 - \left( 1 + \frac{e}{e-1} \cdot \frac{1}{n} \ln(1-\alpha) + O\left(\frac{1}{n^2}\right) \right)^n$$

$$\xrightarrow{n \rightarrow \infty} 1 - e^{\frac{e \ln(1-\alpha)}{e-1}} = 1 - (1-\alpha)^{\frac{e}{e-1}} \neq 1$$

$\Rightarrow$  exp. ne cov.

T-12

$$H_0: p_4 = \frac{8}{24} \quad p_3 = \frac{4}{24} \quad p_2 = p_1 = \frac{6}{24}$$

$$H_1: p_1 = p_2 = p_3 = p_4 = \frac{6}{24}$$

$$\begin{cases} \alpha = 0,2 \\ n = 2 \end{cases}$$

$$\ell(\vec{x}_n) = \frac{\ell_1(\vec{x}_n)}{\ell_0(\vec{x}_n)}$$

$P_0$ :

| $k_0$ | 1              | 2              | 3              | 4              |
|-------|----------------|----------------|----------------|----------------|
| 1     | $\frac{1}{16}$ | $\frac{1}{16}$ | $\frac{1}{24}$ | $\frac{1}{12}$ |
| 2     | $\frac{1}{16}$ | $\frac{1}{16}$ | $\frac{1}{24}$ | $\frac{1}{12}$ |
| 3     | $\frac{1}{24}$ | $\frac{1}{24}$ | $\frac{1}{36}$ | $\frac{1}{18}$ |
| 4     | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{18}$ | $\frac{1}{9}$  |

$$P_{ij} = P_i \cdot P_j$$

$P_1$ :

| $k_1$ | 1              | 2              | 3              | 4              |
|-------|----------------|----------------|----------------|----------------|
| 1     | $\frac{1}{16}$ | $\frac{1}{16}$ | $\frac{1}{16}$ | $\frac{1}{16}$ |
| 2     | $\frac{1}{16}$ | $\frac{1}{16}$ | $\frac{1}{16}$ | $\frac{1}{16}$ |
| 3     | $\frac{1}{16}$ | $\frac{1}{16}$ | $\frac{1}{16}$ | $\frac{1}{16}$ |
| 4     | $\frac{1}{16}$ | $\frac{1}{16}$ | $\frac{1}{16}$ | $\frac{1}{16}$ |

$$P(l \geq C | H_0) \leq 0,2$$

$l$ :

| $l$ | 1             | 2             | 3             | 4              |
|-----|---------------|---------------|---------------|----------------|
| 1   | 1             | 1             | $\frac{3}{2}$ | $\frac{3}{4}$  |
| 2   | 1             | 1             | $\frac{3}{2}$ | $\frac{3}{4}$  |
| 3   | $\frac{3}{2}$ | $\frac{3}{2}$ | $\frac{9}{4}$ | $\frac{9}{8}$  |
| 4   | $\frac{3}{4}$ | $\frac{3}{4}$ | $\frac{9}{8}$ | $\frac{9}{16}$ |

Исходы  $l$  по убыванию:

Самое вероятное для "принятия"  $H_1$ :

- ① (3, 3)
- ② (3, 1) (1, 3) (3, 2) (2, 3)
- ③ (3, 4) (4, 3)

$$\textcircled{1} P_0(3, 3) = \frac{1}{36} \quad \left| \rightarrow \frac{1}{36} + \frac{1}{6} = \frac{7}{36} \approx 0,1944 < \alpha = 0,2$$

$$\textcircled{2} 4 \cdot \frac{1}{24} = \frac{1}{6}$$

$$\textcircled{3} 2 \cdot \frac{1}{18} = \frac{1}{9} \quad \frac{7}{36} + \frac{1}{9} = \frac{11}{36} \approx 0,3 > \alpha$$

$$\Rightarrow \text{перелом} \quad \frac{9}{8} < C \leq \frac{3}{2} \Rightarrow C = \frac{3}{2}$$

$$G: l \geq \frac{3}{2} - \text{крит. обл.}$$

$$W = 5 \cdot \frac{1}{16} = \frac{5}{16} \quad \alpha_1 = \frac{7}{36} \quad \alpha_2 = 1 - W = \frac{11}{16}$$

T-13

Емкость:

$$n = 139$$

$$\sigma_{gr} = 5,722 \text{ мм}$$

$$\sigma_{норм} = 4,612 \text{ мм}$$

$$\alpha = 0,05$$

W-?

Емкость:

$$m = 1000$$

$$\sigma_{gr} = 6,161 \text{ мм}$$

$$\sigma_{норм.} = 3,055 \text{ мм}$$

$$\xi \sim N(\theta_1, \theta_2^2) - \text{группа}$$

$$\eta \sim N(\theta_1', \theta_2'^2) - \text{шум}$$

Данные:

$$1) H_0: \theta_2^2 \text{exp.} = \theta_2^2 \text{er.}$$

$$H_1: \theta_2^2 \text{exp.} \neq \theta_2^2 \text{er.}$$

Шум:

$$2) H_0: \theta_2'^2 \text{exp.} = \theta_2'^2 \text{er.}$$

$$H_1: \theta_2'^2 \text{exp.} \neq \theta_2'^2 \text{er.}$$

$$S^2 = \frac{1}{n-1} \sigma^2$$

$$S_{\text{gr.er.}}^2 = 32,9785$$

$$S_{\text{шум.er.}}^2 = 21,4246$$

$$S_{\text{gr.exp.}}^2 = 37,9959$$

$$S_{\text{шум.exp.}}^2 = 25,5786$$

$$\tilde{\Delta}_{\text{gr.}} = \frac{S_{\text{gr.er.}}^2}{S_{\text{gr.exp.}}^2} = 0,86795$$

$$\tilde{\Delta}_{\text{шум.}} = \frac{S_{\text{шум.er.}}^2}{S_{\text{шум.exp.}}^2} = 0,8376$$

$$\tilde{\Delta} \sim F(n-1, m-1) = F(138, 999)$$

$$\Rightarrow G: \tilde{\Delta} \leq F_{\frac{\alpha}{2}}(138, 999) \quad \text{или} \quad \tilde{\Delta} \geq F_{1-\frac{\alpha}{2}}(138, 999)$$

$$F_{\frac{\alpha}{2}}(138, 999) = 0,76738$$

$$F_{1-\frac{\alpha}{2}}(138, 999) = 1,27169$$

$$\Rightarrow \tilde{\Delta}_{\text{шум.}} \text{ и } \tilde{\Delta}_{\text{gr.}} \notin G \Rightarrow \text{Нельзя отб. } H_0 \text{ где-либо из них}$$

$$W = P(\Delta \in G | H_1) = P(\tilde{\Delta} \leq F_{\frac{\alpha}{2}}(138, 999) | H_1) + \\ + P(\tilde{\Delta} \geq F_{1-\frac{\alpha}{2}}(138, 999) | H_1) =$$

$$= P\left(\frac{S_{\text{er.}}^2}{S_{\text{exp.}}^2} \cdot \frac{\sigma_{\text{exp.}}^2}{\sigma_{\text{er.}}^2} \leq F_{\frac{\alpha}{2}}(138, 999)\right) +$$

$$+ P\left(\frac{S_{\text{er.}}^2}{S_{\text{exp.}}^2} \cdot \frac{\sigma_{\text{exp.}}^2}{\sigma_{\text{er.}}^2} \geq F_{1-\frac{\alpha}{2}}(138, 999)\right) =$$

$$\Rightarrow W = \int_{F_{\frac{\alpha}{2}}(138, 999)/\theta}^{\infty} q(t) dt + \int_0^{F_{1-\frac{\alpha}{2}}(138, 999)/\theta} q(t) dt$$

$\Rightarrow$  расчет в python

T-14

$$\begin{aligned} x &\sim N(a, \sigma_x^2) & \sigma_x^2 &= 2 & \vec{x}_n &= [-1, 14; -6, 10; 2, 42] \\ y &\sim N(b, \sigma_y^2) & \sigma_y^2 &= 1 & \vec{y}_m &= [-2, 29; -2, 91] \end{aligned}$$

$$\alpha = 0,05 \quad n = 3 \quad m = 2$$

$$H_0: a = b$$

no

⊕ Parameter

$$\frac{\bar{x} - a}{\sigma_x} \sqrt{n} \sim N(0, 1)$$

$$H_1: a > b$$

$$\frac{\bar{y} - b}{\sigma_y} \sqrt{m} \sim N(0, 1)$$

$$\left. \begin{aligned} \bar{x} - a &\sim N\left(0, \frac{\sigma_x^2}{n}\right) \\ \bar{y} - b &\sim N\left(0, \frac{\sigma_y^2}{m}\right) \end{aligned} \right\} \Rightarrow \bar{x} - \bar{y} - (a - b) \sim N\left(0, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}\right)$$

$$\Rightarrow \Delta = \frac{\bar{x} - \bar{y} - (a - b)}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim N(0, 1)$$

$$G: \Delta \geq u_{1-\alpha}$$

$$\Delta = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim N(0, 1) \quad ] a - b = 0$$

$$\Rightarrow \bar{x} - \bar{y} \stackrel{H_1}{\sim} N\left(0, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}\right)$$

$$\Rightarrow W = P(\Delta \in G | H_1) = P\left(\Delta \geq u_{1-\alpha}\left(0, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}\right)\right)$$

W(0) python

